1 Solution — GIAM 2.6 Problem 3

1.1 Problem

For each of the following arguments, write it in symbolic form and determine which rules of inference are used.

- *(a)* You are either with us, or you're against us. And you don't appear to be with us. So, that means you're against us!
- *(b)* All those who had cars escaped the flooding. Sandra had a car — therefore, Sandra escaped the flooding.
- (c) When Johnny goes to the casino, he always gambles 'til he goes broke. Today, Johnny has money, so Johnny hasn't been to the casino recently.
- *(d)* (A non-constructive proof that there are irrational a, b with a^b rational.) Either $\sqrt{2}^{\sqrt{2}}$ is rational or it is irrational. If it is rational, let $a = b = \sqrt{2}$. Otherwise let $a = \sqrt{2}^{\sqrt{2}}$ and $b = \sqrt{2}$. In either case there are irrational a, b with a^b rational.

1.2 Answers

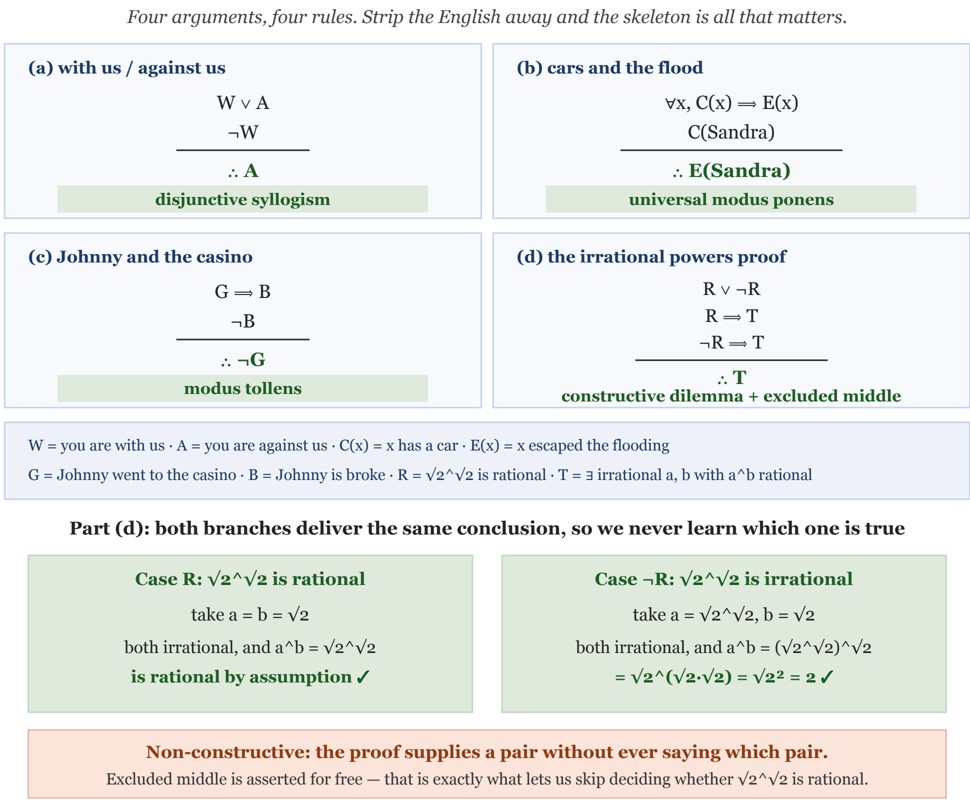

Figure 1.1: Four panels. Panel (a), with us slash against us: premises W or A, and not W, conclusion therefore A — disjunctive syllogism. Panel (b), cars and the flood: premises for-all x C(x) implies E(x), and C(Sandra), conclusion therefore E(Sandra) — universal modus ponens. The third panel, Johnny and the casino: premises G implies B, and not B, conclusion therefore not G — modus tollens. Panel (d), the irrational powers proof: premises R or not-R, R implies T, and not-R implies T, conclusion therefore T — constructive dilemma plus excluded middle. A glossary defines W as you are with us, A as you are against us, C of x as x has a car, E of x as x escaped the flooding, G as Johnny went to the casino, B as Johnny is broke, R as root two to the root two is rational, and T as there exist irrational a and b with a to the b rational. Below, part (d)'s two cases are spelled out. Case R, root two to the root two is rational: take a equals b equals root two; both are irrational and a to the b is rational by assumption. Case not-R, root two to the root two is irrational: take a equals root two to the root two and b equals root two; both are irrational and a to the b equals root two to the power root two times root two, which is root two squared, which is 2. A closing banner notes the proof is non-constructive — it supplies a pair without saying which pair — and that excluded middle is asserted for free, which is exactly what lets us skip deciding whether root two to the root two is rational.

1.2.1 (a) — disjunctive syllogism

Let W = "you are with us" and A = "you are against us."

$$\begin{array}{l} \quad W \vee A \\ \quad \neg W \\ \hline \therefore A \end{array}$$

1.2.2 (b) — universal modus ponens

Let $C(x)$ = "x has a car" and $E(x)$ = "x escaped the flooding."

$$\begin{array}{l} \quad \forall x,\ C(x) \Rightarrow E(x) \\ \quad C(\text{Sandra}) \\ \hline \therefore E(\text{Sandra}) \end{array}$$

This is **universal instantiation** followed by **modus ponens**; at this stage it is fine simply to call it modus ponens and not fuss over the quantifier.

1.2.3 (c) — modus tollens

Let G = "Johnny went to the casino (recently)" and B = "Johnny is broke." "He has money" is $\neg B$.

$$\begin{array}{l} \quad G \Rightarrow B \\ \quad \neg B \\ \hline \therefore \neg G \end{array}$$

1.2.4 (d) — constructive dilemma, with excluded middle

Let R = "$\sqrt{2}^{\sqrt{2}}$ is rational" and T = "there exist irrational a, b with a^b rational."

$$\begin{array}{ll} \quad R \vee \neg R & \text{(law of the excluded middle — free)} \\ \quad R \Rightarrow T & (\text{take } a = b = \sqrt{2}) \\ \quad \neg R \Rightarrow T & (\text{take } a = \sqrt{2}^{\sqrt{2}},\ b = \sqrt{2}) \\ \hline \therefore T & \end{array}$$

All four rules were verified to be tautologies.

1.3 Checking the Two Cases in (d)

Case R. Suppose $\sqrt{2}^{\sqrt{2}}$ is rational. Take $a = b = \sqrt{2}$. Both are irrational (Chapter 1.6), and $a^b = \sqrt{2}^{\sqrt{2}}$ is rational by the case assumption. ✓

Case $\neg R$. Suppose $\sqrt{2}^{\sqrt{2}}$ is irrational. Take $a = \sqrt{2}^{\sqrt{2}}$ and $b = \sqrt{2}$. Both are irrational — a by the case assumption, b by Chapter 1.6 — and

$$a^b = \left(\sqrt{2}^{\sqrt{2}}\right)^{\sqrt{2}} = \sqrt{2}^{\sqrt{2}\cdot\sqrt{2}} = \sqrt{2}^{2} = 2,$$

which is rational. ✓

Both branches deliver T, so T holds. ■

1.4 Remark — Why (d) Is Called Non-Constructive

The proof establishes that a pair (a, b) **exists** without ever telling us which pair it is. It offers two candidates — $(\sqrt{2}, \sqrt{2})$ and $(\sqrt{2}^{\sqrt{2}}, \sqrt{2})$ — and guarantees at least one works, but since we never decide whether $\sqrt{2}^{\sqrt{2}}$ is rational, we never learn which. A **constructive** proof would hand over a specific witness and verify it. (An easy constructive alternative exists: $a = \sqrt{2}$, $b = 2\log_2 3$ gives $a^b = 3$. But the point of the exercise is the *shape* of the argument, not the sharpest result.)

The engine of the non-constructiveness is the very first line: **excluded middle**, $R \vee \neg R$, asserted for free exactly as in Problem 2. That single free premise is what buys us permission to reason about both cases without resolving either.

1.5 Remark — As It Happens, We *Do* Know Which Case Holds

By the **Gelfond–Schneider theorem** (1934), α^β is transcendental whenever α is algebraic and $\neq 0, 1$ and β is algebraic and irrational. Taking $\alpha = \beta = \sqrt{2}$ shows $\sqrt{2}^{\sqrt{2}}$ is transcendental, hence **irrational** — so Case $\neg R$ is the real one, and the honest witnesses are $a = \sqrt{2}^{\sqrt{2}}$, $b = \sqrt{2}$.

But notice: the proof in part (d) does not need, use, or even mention this. Gelfond–Schneider is a deep theorem; the pub-style case split is elementary. That is the appeal of non-constructive arguments — they can settle an existence question far more cheaply than any method that must actually produce the object.

1.6 Remark — Modus Tollens Is Not "Denying the Antecedent"

Part (c) is the one most often gotten wrong. The valid move is

$$\underbrace{G \Rightarrow B,\ \neg B \ \vdash\ \neg G}_{\text{modus tollens — valid}} \qquad \text{not} \qquad \underbrace{G \Rightarrow B,\ \neg G \ \vdash\ \neg B}_{\text{denying the antecedent — invalid}} \ .$$

Johnny having money tells us the *consequent* failed, so the antecedent must have failed too — you may run the arrow backwards only when you negate. Had the argument instead said "Johnny hasn't been to the casino, so he must have money," it would be fallacious: he could be broke for entirely unrelated reasons. Modus tollens is just the contrapositive of Problem 2.2.8 combined with modus ponens: from $G \Rightarrow B$ get $\neg B \Rightarrow \neg G$, then apply $\neg B$.

1.7 Remark — Part (a)'s Rhetoric vs. Its Logic

The form of (a) is valid, but the argument is a textbook piece of political rhetoric and repays a second look. Its first premise, $W \vee A$, is a **false dichotomy** unless "against us" is *defined* as "not with us" — neutrality, indifference, and partial agreement are all excluded by fiat. This is exactly the trap identified in Problem 2: a disjunction that *looks* like a free tautology $P \vee \neg P$, but is doing real (and contestable) work. Note too the slide in the second premise — the speaker says "you don't *appear* to be with us" and then treats it as $\neg W$. Disjunctive syllogism is perfectly valid; the mischief is entirely in what gets fed to it.

1.8 Remark — The Four Rules Side by Side

Collecting the patterns for reference:

part	**rule**	**premises**	**conclusion**
(a)	disjunctive syllogism	$P \vee Q$, $\neg P$	Q
(b)	(universal) modus ponens	$P \Rightarrow Q$, P	Q
(c)	modus tollens	$P \Rightarrow Q$, $\neg Q$	$\neg P$
(d)	constructive dilemma	$P \vee Q$, $P \Rightarrow R$, $Q \Rightarrow R$	R

Table 1.1

Together with hypothetical syllogism from Problem 1, these five account for the overwhelming majority of steps in ordinary mathematical writing. Learning to spot them under the English — as all three problems in this section have asked — is the point: once the skeleton is visible, checking validity is mechanical, and all your attention can go to whether the premises deserve to be believed.